MEDIUM · ALGEBRA

Linear functions

03

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 GRID-IN ALGEBRA

$$c(x) = mx + 500$$

A company's total cost $c(x)$, in dollars, to produce x shirts is given by the function above, where m is a constant and $x > 0$. The total cost to produce 100 shirts is \$800. What is the total cost, in dollars, to produce 1000 shirts? (Disregard the \$ sign when gridding your answer.)

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q2 ALGEBRA

The relationship between two variables, x and y, is linear. For every increase in the value of x by 1, the value of y increases by 8. When the value of x is 2, the value of y is 18. Which equation represents this relationship?

A. $y = 2x + 18$

B. $y = 2x + 8$

C. $y = 8x + 2$

D. $y = 3x + 26$

Q3 ALGEBRA

The function h is defined by $hx = 4x + 28$. The graph of $y = hx$ in the xy-plane has an x-intercept at $a, 0$ and a y-intercept at $0, b$, where a and b are constants. What is the value of $a + b$?

A. 21

B. 28

C. 32

D. 35

Q4 ALGEBRA

$$w(t) = 300 - 4t$$

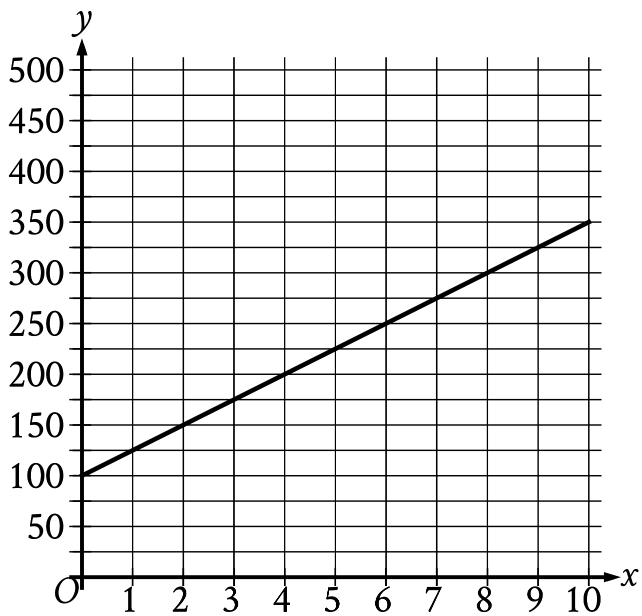
The function w models the volume of liquid, in milliliters, in a container t seconds after it begins draining from a hole at the bottom. According to the model, what is the predicted volume, in milliliters, draining from the container each second?

A. 300

B. 296

C. 75

D. 4



The graph of the function f, where $y = f(x)$, gives the total cost y, in dollars, for a certain video game system and x games. What is the best interpretation of the slope of the graph in this context?

A. Each game costs \$25.

B. The video game system costs \$100.

C. The video game system costs \$25.

D. Each game costs \$100.

Q6

ALGEBRA

A team of workers has been moving cargo off of a ship. The equation below models the approximate number of tons of cargo, y, that remains to be moved x hours after the team started working.

$$y = 120 - 25x$$

The graph of this equation in the xy-plane is a line. What is the best interpretation of the x-intercept in this context?

A. The team will have moved all the cargo in about 4.8 hours.

B. The team has been moving about 4.8 tons of cargo per hour.

C. The team has been moving about 25 tons of cargo per hour.

D. The team started with 120 tons of cargo to move.

Q7

ALGEBRA

If $f(x) = x + 6$ and $g(x) = 6x$, what is the value of $7f(2) - g(2)$?

A. -4

B. 8

C. 42

D. 44

Q8 ALGEBRA

For the linear function h, the graph of $y = hx$ in the xy-plane passes through the points $7, 21$ and $9, 25$. Which equation defines h?

A. $hx = \frac{1}{2}x - \frac{7}{2}$

B. $hx = 2x + 7$

C. $hx = 7x + 21$

D. $hx = 9x + 25$

Q9 ALGEBRA

The table gives the number of hours, h, of labor and a plumber's total charge fh, in dollars, for two different jobs.

h	fh
1	155
3	285

There is a linear relationship between h and fh. Which equation represents this relationship?

A. $fh = 25h + 130$

B. $fh = 130h + 25$

C. $fh = 65h + 90$

D. $fh = 90h + 65$

Q10

ALGEBRA

In the xy-plane, the graph of the linear function f contains the points $0, 3$ and $7, 31$. Which equation defines f, where $y = fx$?

A. $fx = 28x + 34$

B. $fx = 3x + 38$

C. $fx = 4x + 3$

D. $fx = 7x + 3$